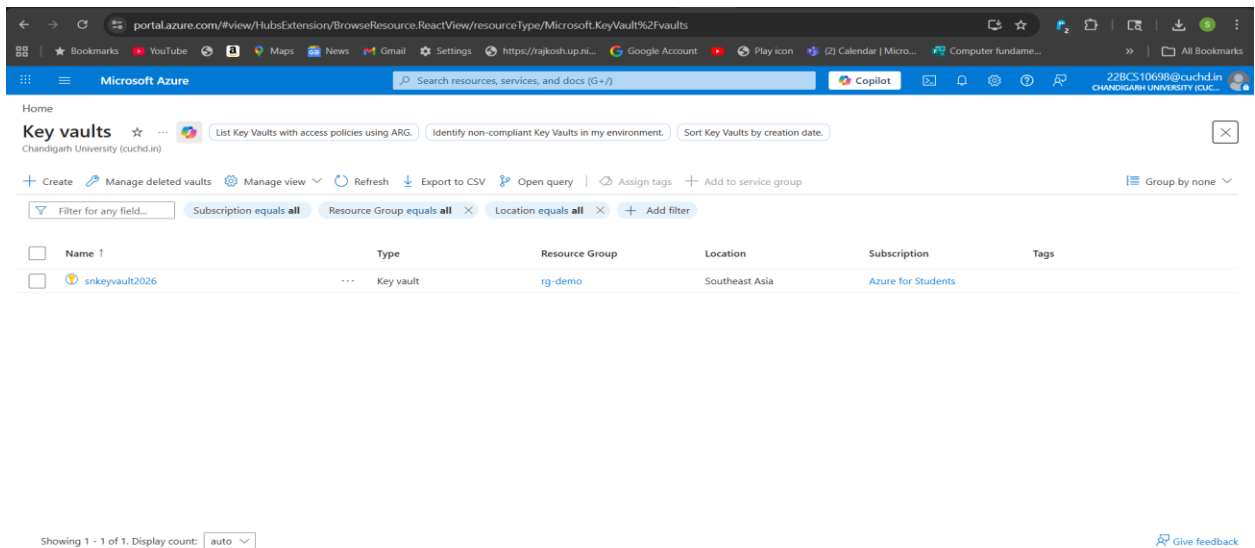


Azure Key Vault (Secrets and Keys)

This document explains Azure Key Vault creation and usage of Keys/Secrets with a .NET Console Application.

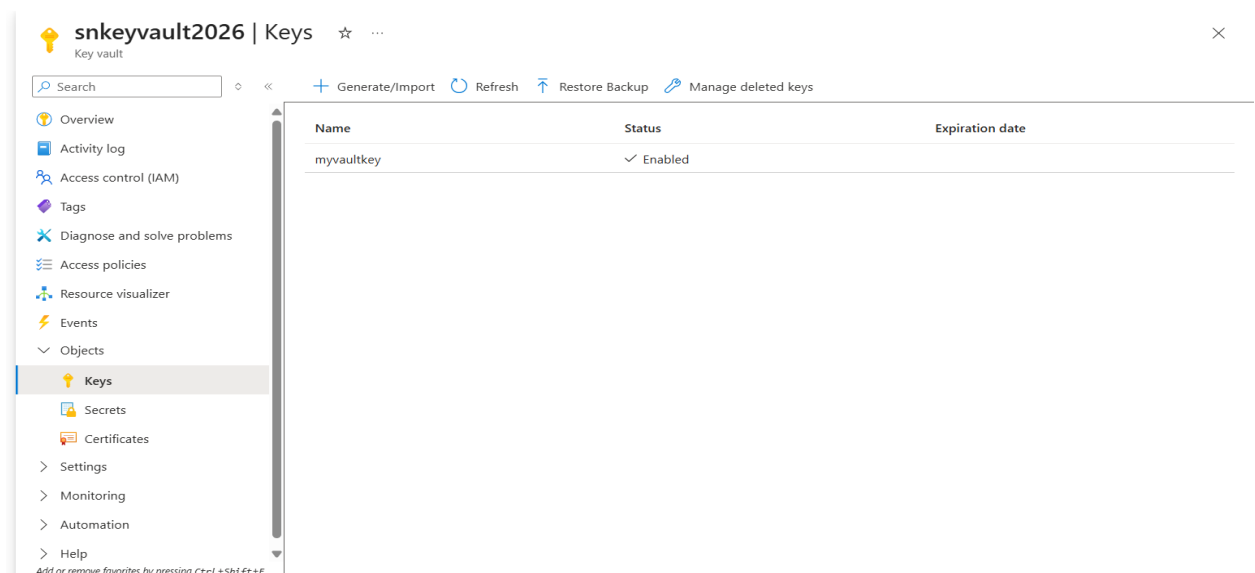
Step 1: Create Azure Key Vault

Open Azure Portal and verify/create the Key Vault resource.



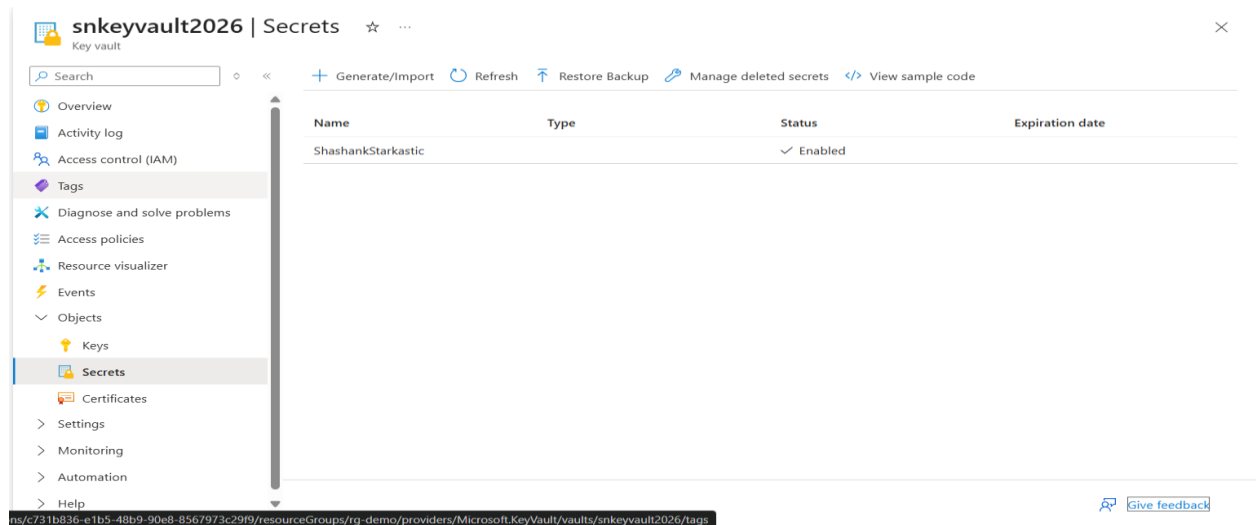
Step 2: Create Key in Vault

Inside the Key Vault, open Keys and create/import the encryption key.



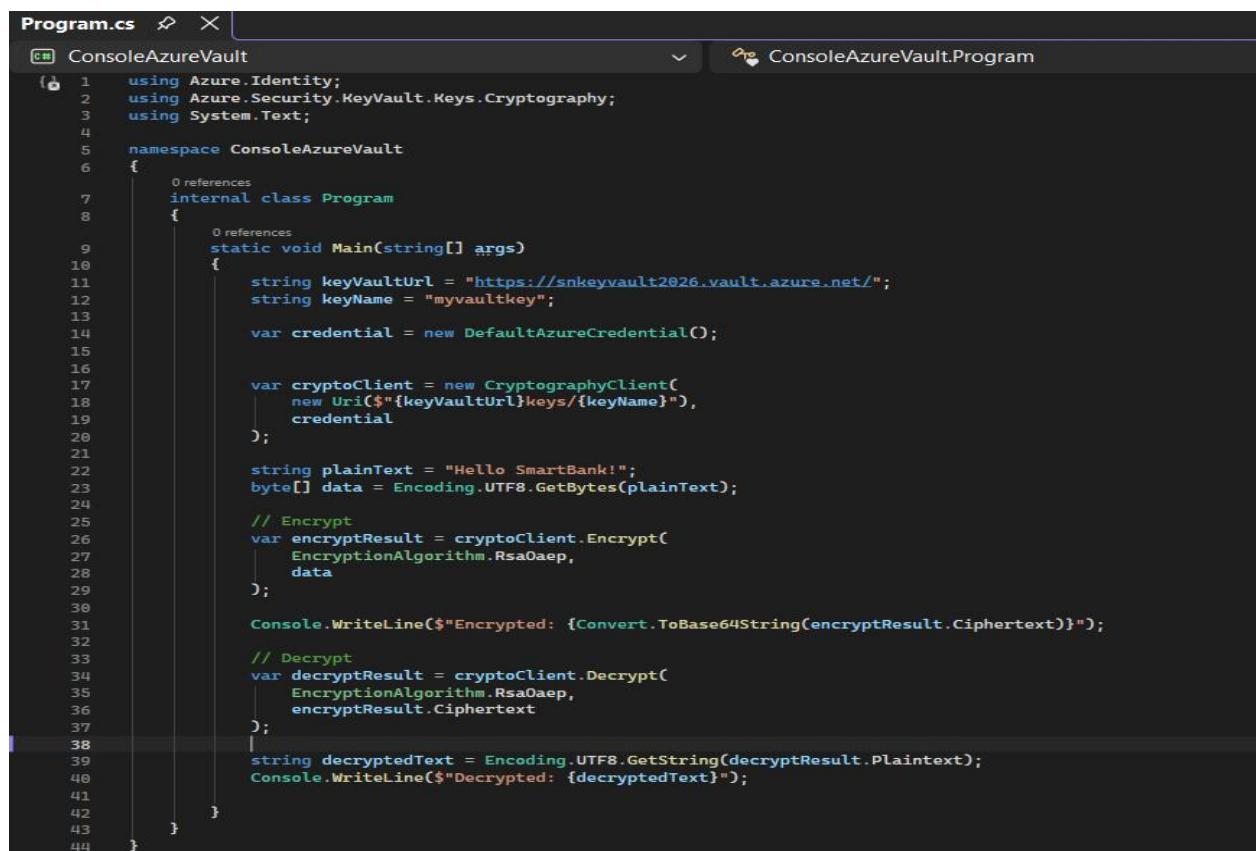
Step 3: Create Secret in Vault

Open Secrets and create a secret value to store sensitive data securely.



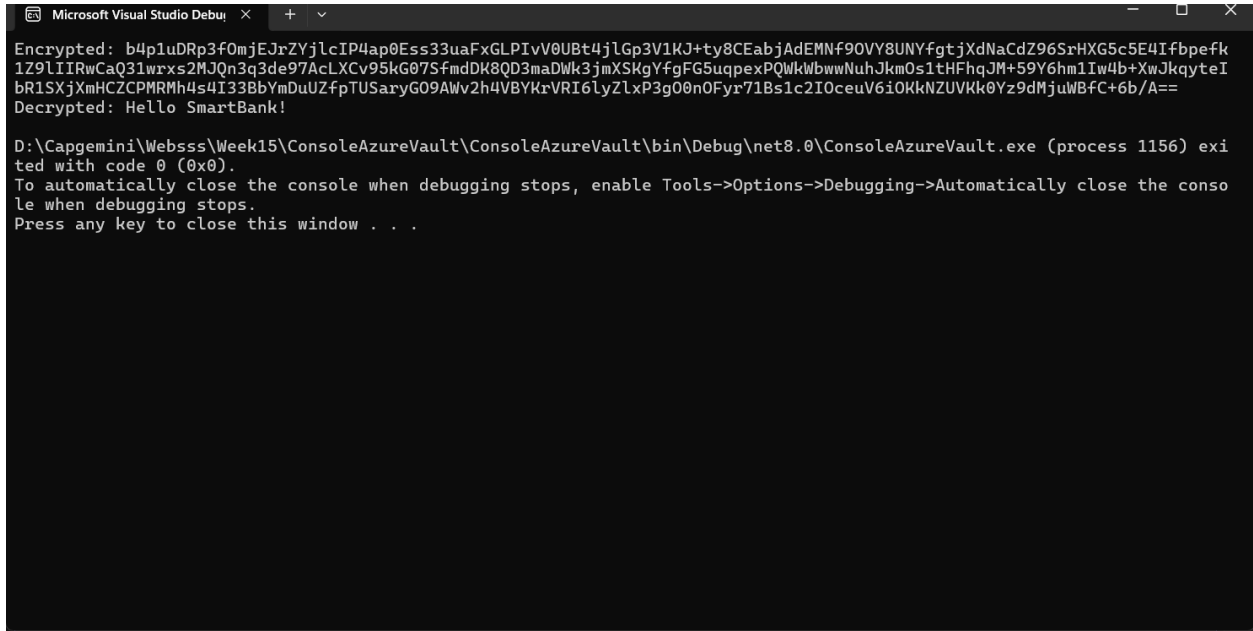
Step 4: Use Key Vault in Console App

In Visual Studio, write C# code to connect to Key Vault and perform encryption/decryption using CryptographyClient.



Step 5: Run Application and Verify Output

Execute the console app. Encrypted text is generated and then decrypted back to original text successfully.



```
Microsoft Visual Studio Debug Console
Encrypted: b4p1uDRp3f0mjEJrZYjlcIP4ap0Ess33uaFxGLPIvV0UBt4jLGP3V1KJ+ty8CEabjAdEMNF90VY8UNYfgtjXdNaCdZ96SrHXG5c5E4Ifbpefk
1Z9LIIRwCaQ31wrxs2MJQn3q3de97AcLXCv95kG07SfmdDK8QD3maDwk3jmXSKgYfgFG5uqpexPQWkwBwwNuhJkm0s1tHFhqJM+59Y6hm1Iw4b+XwJkqyteI
bR1SXjXmHCZCPMRMh4s4I33BbYmDuUZfpTUSaryG09AWv2h4VBYKrVRI6lyZLxP3g00n0Fyr71Bs1c2IOceuV6i0KkNZUVKk0Yz9dMjuWBfC+6b/A==
Decrypted: Hello SmartBank!

D:\Capgemini\Websss\Week15\ConsoleAzureVault\ConsoleAzureVault\bin\Debug\net8.0\ConsoleAzureVault.exe (process 1156) exi
ted with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the conso
le when debugging stops.
Press any key to close this window . . .
```

Conclusion

Azure Key Vault securely stores secrets and keys, while applications can safely use them for encryption and sensitive configuration management.